

Mini "Data Center" Water Purification Project

Concept: ESP32 controller (data center ka "brain") + heater (waste heat source) se paani ko distill/purify karna — hackathon-style project.

Yeh Kaam Kaise Karta Hai

- Heating** — Heater coil paani ko garam karta hai (represents "data center ki waste heat")
- Evaporation** — Garam hone se paani se vapor uthta hai — salt/impurities peeche reh jaati hain, sirf pure H₂O evaporate hota hai
- Condensation** — Vapor upar uthke thandi angled lid se touch karta hai, wapas liquid ban jaata hai
- Collection** — Tilt ki wajah se clean paani side mein slide karke spout se collection cup mein girta hai
- Control** — ESP32 + Relay heater ko ON/OFF karte hain; temp sensor overheating se bachata hai

Kya Problem Solve Kar Raha Hai

Real AI data centers bohot zyada **electricity aur paani** consume karte hain — cooling ke liye paani use hota hai (waste/contaminate ho jaata hai), aur cooling process khud bhi heat generate karta hai jo simply discard ho jaati hai.

Iska idea: **waste heat ko reuse karo** paani purify karne ke liye — taaki:

- Dirty/salty paani drinkable bane
- Discard hone wali heat productive kaam kare
- Ek hi setup mein cooling + clean water dono ho ("two birds, one stone")

Equipment List, Cost & Kaam

#	Equipment	Cost (₹)	Kaam (Purpose)
1	ESP32 Dev Board	250–400	Controller — relay ko ON/OFF karta hai, temp sensor read karta hai, WiFi se data bhejta hai
2	Heating element (nichrome wire coil / 12V DC heater cartridge, ~20-40W)	50–200	Paani ko evaporate karne ke liye heat generate karta hai — yeh "data center ki waste heat" hai
3	Relay module (5V, 1-channel)	80–120	ESP32 (low power) ko heater (high power) safely switch karne deta hai
4	DS18B20 waterproof temp sensor	100–150	Chamber/water ka temperature measure karta hai, ESP32 ko data deta hai
5	4.7kΩ resistor (pull-up)	5–10	DS18B20 sensor ko sahi se kaam karne ke liye zaroori

6	Airtight glass/plastic container (lower chamber)	100–200	Feed water (dirty/salty paani) rakhne ke liye — yahi evaporate hoga
7	Clear glass/acrylic sheet (angled lid)	0–100	Condensation surface — vapor isi pe thanda hoke paani banta hai
8	Collection cup	0	Clean/distilled paani collect karne ke liye (ghar pe hi milega)
9	Silicone sealant	100	Chamber ko airtight seal karne ke liye — leak na ho
10	Jumper wires + hookup wire	100–150	Sab components ko connect karne ke liye
11	Breadboard/perfboard	80–150	Circuit assemble karne ke liye
12	12V power adapter/battery pack	150–300	Heater ko power dene ke liye
13	USB cable (ESP32 programming/power)	100–150	ESP32 ko power + code upload karne ke liye

Total estimated cost: ₹1,400 – ₹1,900 (agar sab naya lena pade) (*agar breadboard, wires, ya container ghar pe already hain toh aur kam ho jayega*)

Kaise Banana Hai — Step by Step

Step 1: Chamber Taiyar Karo

Lower container lo — yeh feed water chamber banega. Isme dirty/salty paani daalna hai jo purify hoga.

Step 2: Heater Fix Karo

Nichrome wire coil ya DC heating element ko container ke andar/neeche fix karo (waterproof insulation ke saath — bare wires paani ko touch nahi karni chahiye directly, warna short circuit).

Step 3: Relay Wiring

- Relay ka control pin → ESP32 GPIO pin
- Relay ka power side → heater + 12V adapter ke beech mein laga do
- Isse ESP32 heater ko ON/OFF control kar payega

Step 4: Temp Sensor Lagao

DS18B20 sensor ko water chamber ke paas (par electronics se door) fix karo. Data pin ko ESP32 ke GPIO se connect karo, 4.7kΩ resistor pull-up ke liye lagao.

Step 5: Chamber Seal Karo

Container ke upar angled clear lid lagao (thoda tilt kiya hua — taaki condensed drops ek side slide karke gire). Silicone sealant se sab edges seal karo — airtight hona zaroori hai.

Step 6: Collection Point

Lid ke sabse neeche/tilted point ke neeche ek chhota cup rakho jahan clean paani drip karke jama hoga.

Step 7: Code Upload Karo

Arduino IDE (ya MicroPython) se ESP32 pe ek simple script upload karo:

- Relay ko ON karo (heater start)
- Temperature continuously read + log karo
- (Optional) WiFi se data bhejo apne phone/laptop dashboard pe

Step 8: Test Run

Thoda sa paani (saltwater/dirty water) daalo chamber mein, system ON karo, aur dekho kitni der mein kitna clean paani collect hota hai. Pehli baar chhoti quantity se start karo.

Step 9: Safety Check

- Saari electronics (ESP32, relay, wires) paani se completely alag/sealed rakho
- Heater ke wires ache se insulated hone chahiye
- Loose connections double-check karo before power ON karna

Baaki Bachi Heat Ka Kya Karein

Heater se extra heat:

- **Preheat loop:** Excess heat se agle batch ka fresh feed water preheat karo — next cycle mein kam energy lagegi (real desalination plants yehi karte hain — "multi-effect distillation")
- **Insulation:** Chamber ko bahar se thermocol/foam se wrap karo — heat waste kam hogi, evaporation mein zyada use hogi
- **Safety vent:** Thoda exhaust vent lagao jo excess heat ko safely bahar nikale, electronics se door — overheating se bachne ke liye

Condensation se nikalti heat (lid pe):

- Vapor jab lid pe condense hota hai, tab bhi heat release hoti hai (latent heat) — yeh ignore mat karna
- **Simplest fix:** Lid ko glass ke bajaye **metal** ka banao, bahar side chhote fins/heatsink lagao — heat jaldi bahar nikalegi, condensation bhi fast hoga (double benefit)
- **Upgrade (optional):** Lid ke peeche ek thin metal tube route karo jisme agla fresh feed water flow kare — condensation heat usse preheat kar degi (professional "multi-effect" touch, interview mein bolne layak)
- **Basic option:** Ek chhota fan lid ke bahar laga do — condensation fast + heat disperse, sabse easy add-on

Recommended start: Metal lid + fins (simple, kam components) — baad mein preheat loop add kar sakta hai as upgrade.

Note

Yeh chhoti scale ka demo/proof-of-concept hai (jaise hackathon mein bhi tha — sirf kuch ml paani banega). Goal hai concept dikhana, industrial-scale purification nahi.

